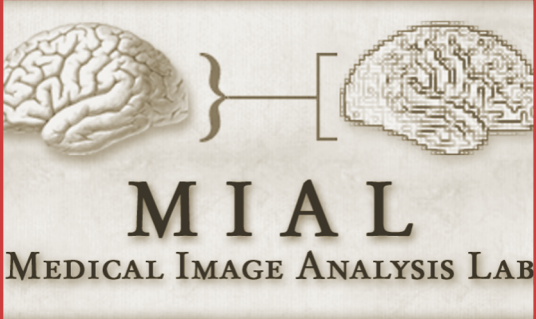


Clinical User-Centered Explainable AI for Medical Image Analysis

A Critical Technical Practice



Weina Jin, MD

Medical Image Analysis Lab, School of Computing Science, Simon Fraser University, Canada

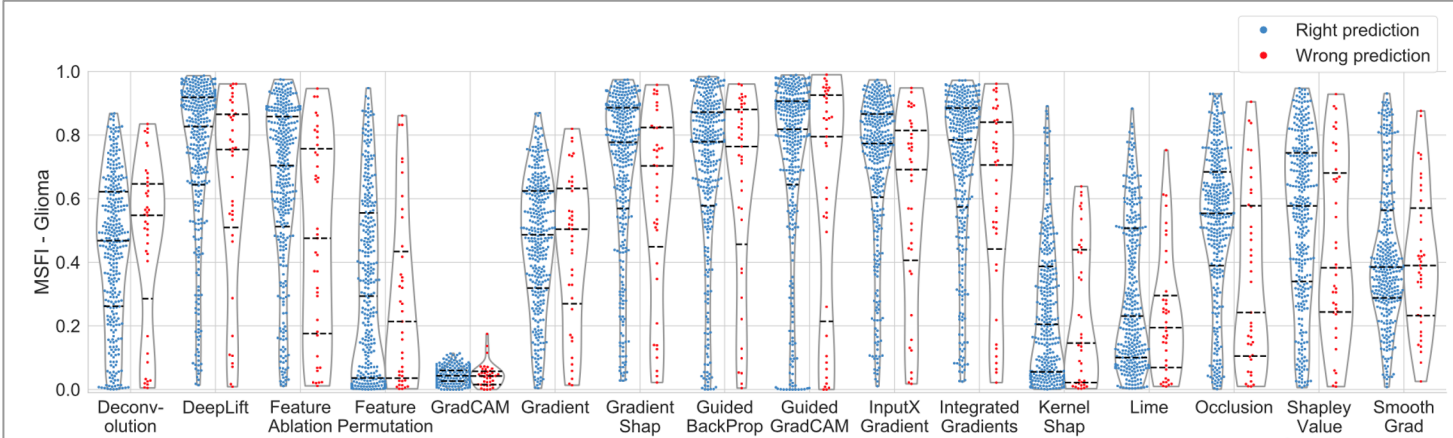
SFU

Thesis Research Question

How can we—both as the collective technical community and as the individual technical developers and researchers—develop **explainable AI (XAI)** and **medical image analysis** techniques **responsibly** to genuinely benefit patients' healthcare and society?

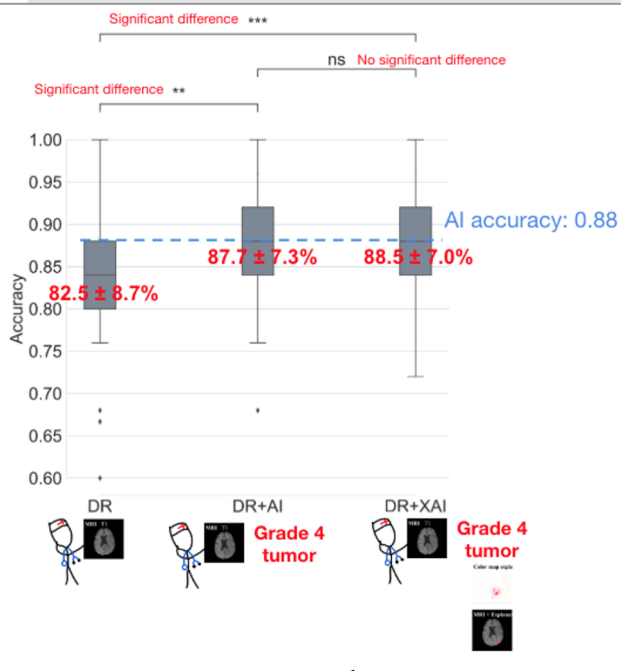
Symptoms

Existing XAI do not fulfill clinical requirements



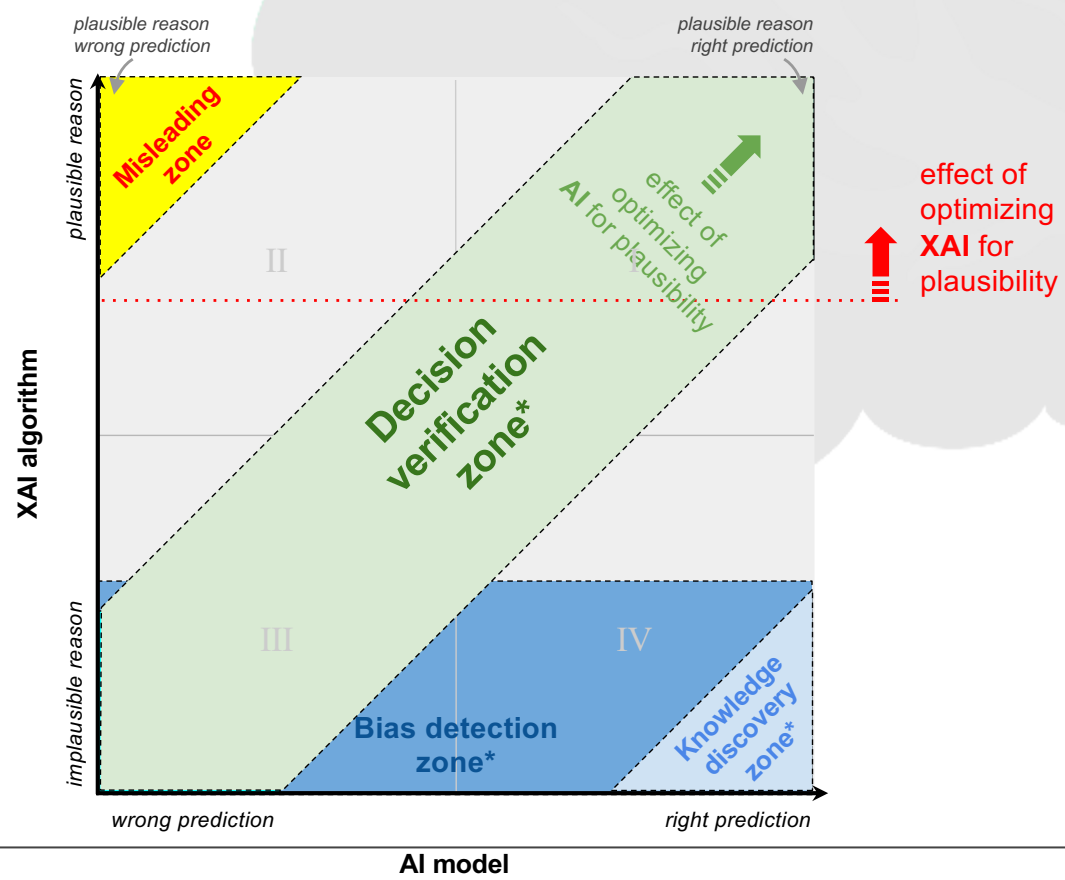
W. Jin, X. Li, G. Hamarneh, "Evaluating Explainable AI on Multi-Modal Medical Imaging Task: Can Existing Algorithms Fulfill Clinical Requirements?" AAAI 2022

Existing XAI is not helpful to improve doctors' task performance



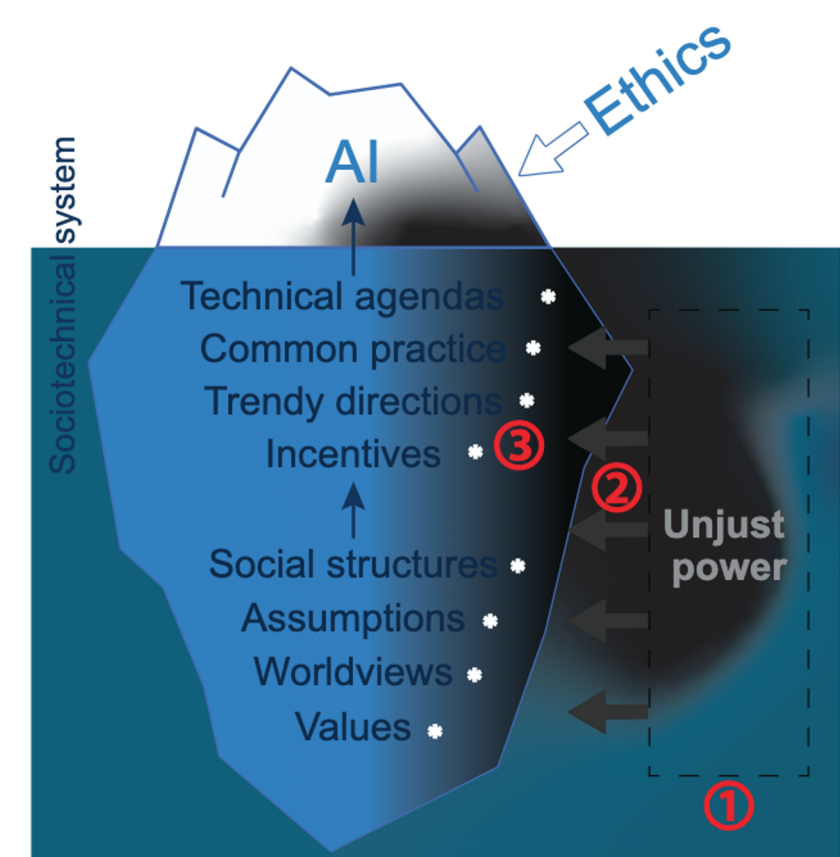
W. Jin (co-first author), M. Fatehi (co-first author), R. Guo, G. Hamarneh, "Evaluating the Clinical Utility of Artificial Intelligence Assistance and its Explanation on the Glioma Grading Task". Artificial Intelligence in Medicine. 2024

The main XAI evaluation criterion of plausibility is unscientific and unethical



Diagnosis

The root causes of the symptoms are the imbalanced power between for-profit sector and users/the public



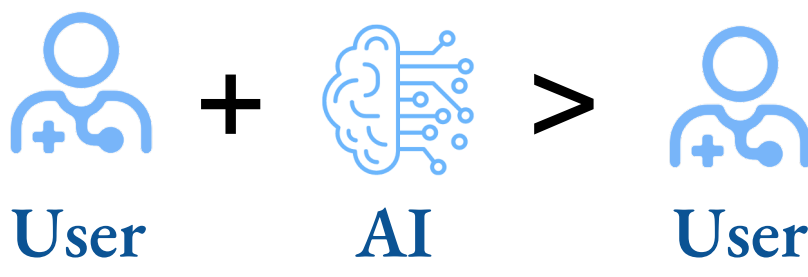
Interventions

critical technical practice

Critical analysis of AI-assisted user studies

Over-reliance can explain away the performance gain

Is the performance of User+AI better than User alone?



The AI assistance is considered to show a positive effect by helping users improve their performance

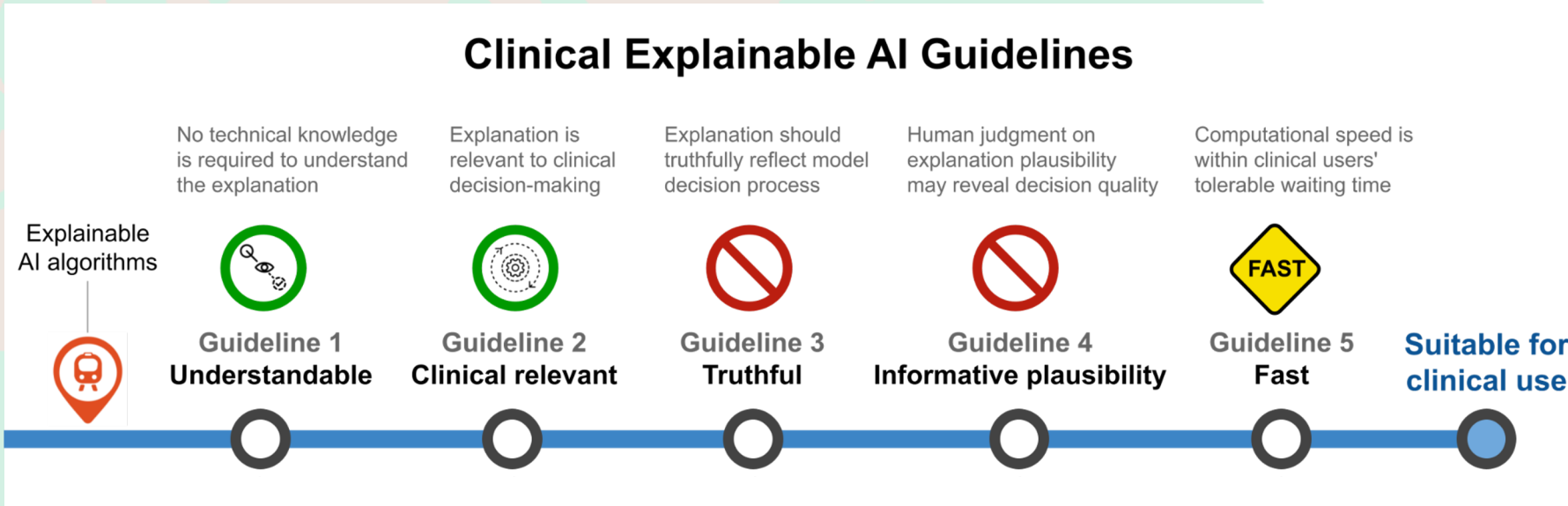
FLAWED

W. Jin, G. Hamarneh, "Accounting for Over-Reliance in AI-Assisted Performance Studies". MICCAI Workshop on Human-AI Collaboration. 2026

Responsible XAI framework

1. Basic properties of XAI P1: Power check. XAI is to check and shift power from the profit-oriented sector to users and society, especially the marginalized and mostly impacted communities. P2: User-centred XAI. XAI should support users' intended goals of using AI explanation and users' reasoning process. P3: Bounded functionality. The XAI/AI system has scopes, weaknesses, and limitations, some of which cannot be overcome by technical advances. P4: Rejectable AI. The associated AI system of the XAI technique and its data collection are rejectable. P5: Unnecessary trade-off. Explainability does not inevitably trade off with performance.	3. XAI evaluation XAI algorithmic efficacy: requiring the explanation <i>content</i> to be truthful and useful <i>Truthfulness</i> is the primary evaluation objective for XAI. <i>Users' intended purposes</i> are the secondary evaluation objective for XAI. XAI techniques can be optimized and evaluated for a specific functionality, such as verifying AI prediction, detecting bias from AI prediction, and discovering new knowledge. Limitation analysis is as important as efficacy evaluation. It includes three components: L1. Systematically identify and assess the scope, limitations, weaknesses, failure modes, negative impacts, and risks of the XAI technique and its coupled AI system. - Quantify the probability of misleading explanations from the AI-XAI system on the given task. L2. Disclose technique-specific limitations of the XAI technique and its evaluation methods. [When applicable] Declare the limitation of using approximation methods (either computationally or with human subjects) to the gold standard assessment to assess the utility of XAI to users' target task. L3. Disclose the intrinsic limits of the XAI technique and its evaluation methods. [When applicable] Acknowledge the intrinsic limit of truthfulness evaluation that can only approximate, but not fully access, the underlying prediction process of a complex black-box model.
2. XAI design: requiring the explanation <i>form</i> to be understandable and fully explainable The explanation form of an XAI technique needs to be able to support each step in users' intuitive inferential reasoning of an AI explanation by providing explanatory information that can fully address the three propositions: ① How are the important features <i>A</i> identified from the input instance <i>X</i>? ② How is the important features <i>A</i> described and presented in human-understandable forms? ③ How is the output prediction <i>Y</i> inferred from the important features <i>A</i>?	

Clinical XAI guidelines



W. Jin, X. Li, M. Fatehi, G. Hamarneh, "Guidelines and Evaluation of Clinical Explainable AI in Medical Image Analysis". Medical Image Analysis 2023

Ethical Medical Image Synthesis

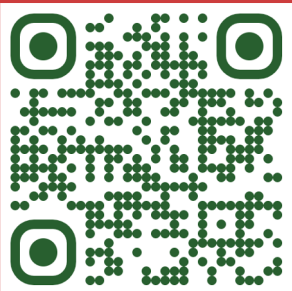
End-user-centered XAI framework EUCA

Avoid misinformation Terminology - Use terms that can clearly distinguish synthetic from medical images. Technical standard - Label synthetic images in all outputs, including report, code, interface, etc. Usage declaration - Declare the usage of synthetic images in tasks, datasets, and models.	Justifiable motivation Benefit whom? - Do the motivations stem from clinical stakeholders' and public's interests? Provide evidence - Are the motivations justifiable by scientific and ethical evidences? Who are at risk? - Are the motivations balanced by emphasizing who may be at risk?	Design assumption Clinical knowledge - Detail design assumptions and the incorporated clinical knowledge. Distribution shift - Detail what potential data distribution shifts the model introduces. Bias - Detail what potential biases the model introduces.	Evaluation Realism - The realism evaluation should be grounded in real medical phenomena. Downstream task - The benefit claims should not go beyond the evaluation scope. Avoid overclaiming - Avoid unscientific claims on the efficacy and benefits of the model.	Limitation analysis Limitation assessment - Conduct thorough assessments on the negative aspects of model. Technical limitations - Acknowledge technique-specific limitations in design and evaluation. Intrinsic limits - Acknowledge the intrinsic limits of synthetic data to avoid ignoring its harms.
---	--	--	--	---

Category	Explanation Forms and Examples
Contextual Information	Input, Output, Dataset, Performance
Features	Feature Attribution, Feature Shapes, Feature Interaction
Examples	Similar Examples, Typical Examples, Counterfactual Examples
Rules	Decision Rule, Decision Tree

Encoding justice as foundational assumptions in technical development

Mainstream imagination of AI & its questionable assumptions (Sec. 2) S1. AI can be capable of solving many hard problems - It surreptitiously substitutes model for the real, and ignores AI's fundamental dependencies on human experience and natural resources. S2. AI replacement can free humans - It forgets AI's fundamental dependence on labor, devalues workers and degrades work to make AI appear "automatic." S3. AI can improve productivity - It is more likely to lead to inequality, as it inherently lacks power check mechanism to ensure shared prosperity.	Demonstrating our process of constructing a new imagination: AI for just work (Sec. 4) Baking ethics into the foundational assumptions (Sec. 4.1) A1. AI has intrinsic limits and fundamental dependencies A2. Worker and work are not devalued or degraded A3. Prioritize ethics over productivity Deducing AI properties from assumptions (Sec. 4.2) P1. AI is grounded in real-world problems P2. Thorough limitation analysis of AI is a default P3. AI and data collection are rejectable P4. Choose to design AI that can increase workers' power Applying the new imagination in AI technical development (Sec. 4.3) - Propose ethical criteria and evaluation paradigm to ground AI in real-world problems - Develop three aspects of limitation analysis to assess and declare AI's limitations - Develop a stakeholders' checklist to question the appropriateness of AI
--	--



PhD Thesis Webpage
weina.me/phd_thesis



Video Presentation
<https://youtu.be/bIUyLrWfIXc>